



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Volume & Surface Area

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Your notes

Volume

Volume

What is volume?

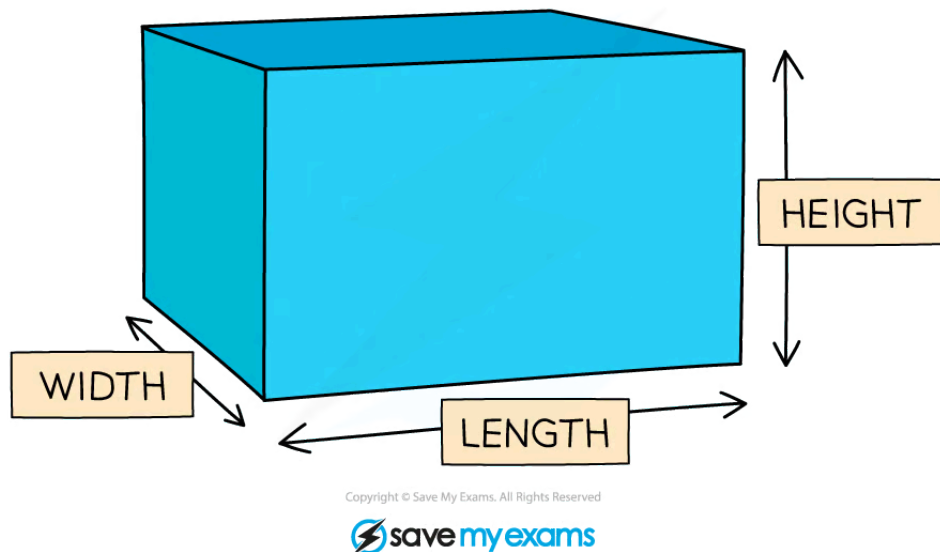
- The **volume** of a 3D shape is a measure of how much **space** it takes up
- You need to be able to calculate the volumes of a number of **common 3D shapes**, including:
 - Cubes and cuboids
 - Prisms
 - Pyramids
 - Cylinders
 - Spheres

How do I find the volume of a cube or a cuboid?

- A **cube** is a special **cuboid**, where the length, width and height are all of **equal length**
- A **cuboid** is another name for a rectangular-based **prism**
- To find the volume, V , of a **cube** or a **cuboid**, with length, l , width, w , and height, h , use the formula
 - $V = lwh$
 - This formula is **not** given to you in the exam



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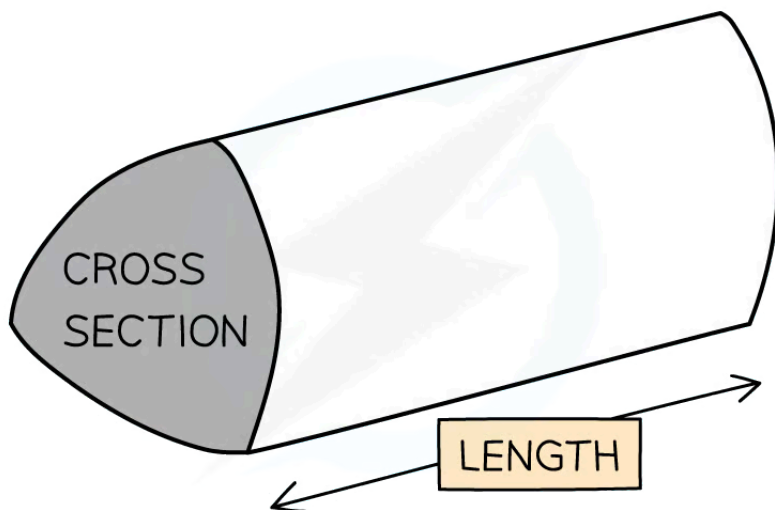
- You will sometimes see the terms 'depth' or 'breadth' instead of 'height' or 'width'

How do I find the volume of a prism?

- A **prism** is a 3D object with a **constant cross-sectional area**
- To find the volume, V , of a **prism**, with cross-sectional area, A , and length, l , use the formula
 - $V = Al$
 - This formula is given to you in the exam



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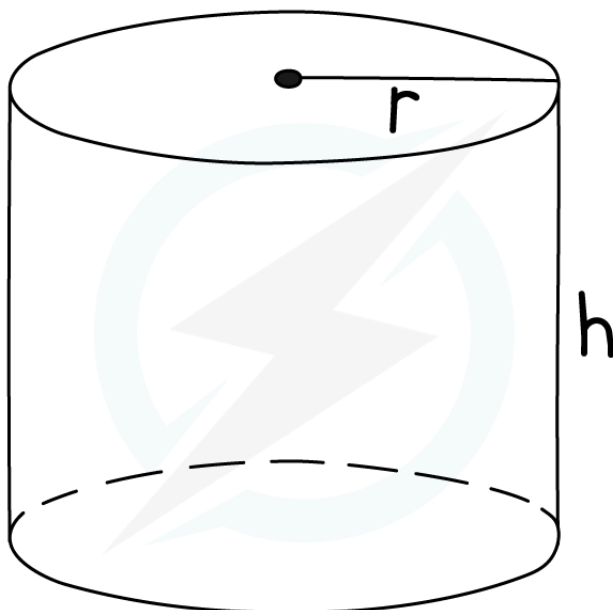
- Note that the cross-section can be **any shape**, so as long as you know its **area** and the **length** of the prism, you can calculate its **volume**
 - If you know the **volume** and **length** of the prism, you can calculate the **area** of the **cross-section**

How do I find the volume of a cylinder?

- To calculate the volume, V , of a **cylinder** with radius, r , and height, h , use the formula
 - $V = \pi r^2 h$
 - This formula is given to you in the exam



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- Note that a cylinder is similar to a **prism**, its cross-section is a circle with area πr^2 , and its length is h

How do I find the volume of a pyramid?

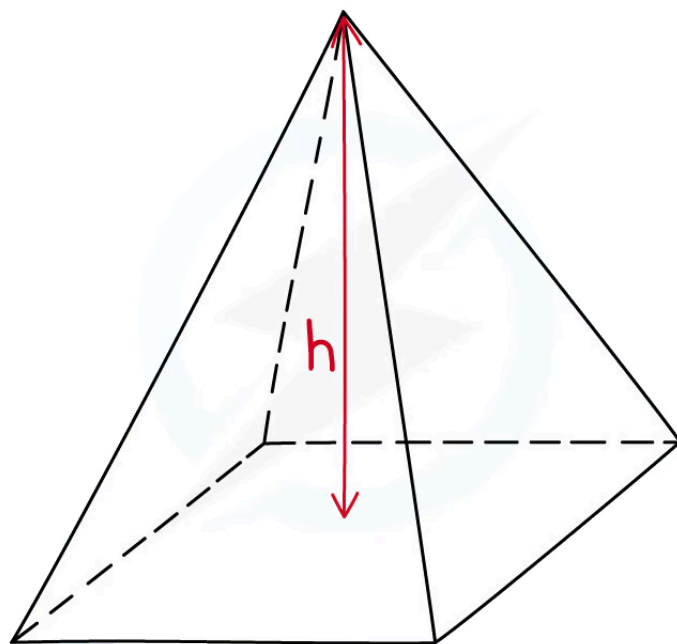
- To calculate the volume, V , of a **pyramid** with base area, A , and perpendicular height, h , use the formula

- $$V = \frac{1}{3} Ah$$

- This formula is given to you in the exam



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- The height must be a line from the top of the pyramid that is **perpendicular** to the base
- The **base** of a pyramid could be a square, a rectangle or a triangle

How do I find the volume of a cone?

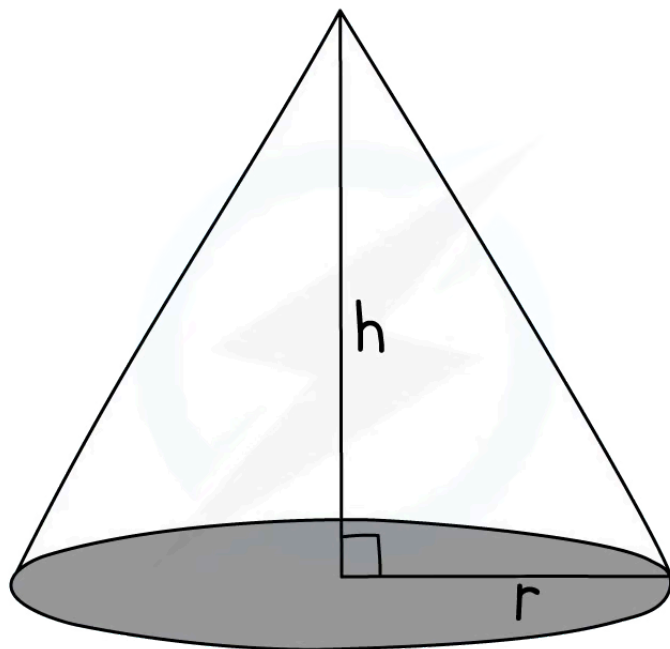
- To calculate the volume, V , of a **cone** with base radius, r , and perpendicular height, h , use the formula

- $$V = \frac{1}{3} \pi r^2 h$$

- This formula is given to you in the exam



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- Note that volume formula for a **cone** is similar to a **pyramid**
- The height must be a line from the top of the cone that is **perpendicular** to the base

How do I find the volume of a sphere?

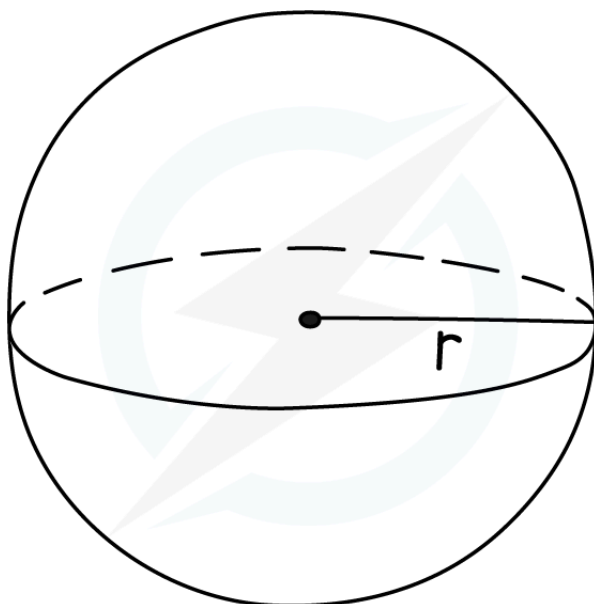
- To calculate the volume, V , of a **sphere** with radius, r , use the formula

- $V = \frac{4}{3} \pi r^3$

- This formula is given to you in your exam



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Examiner Tips and Tricks

- You only need to memorise the volume formulae for **cuboids**!

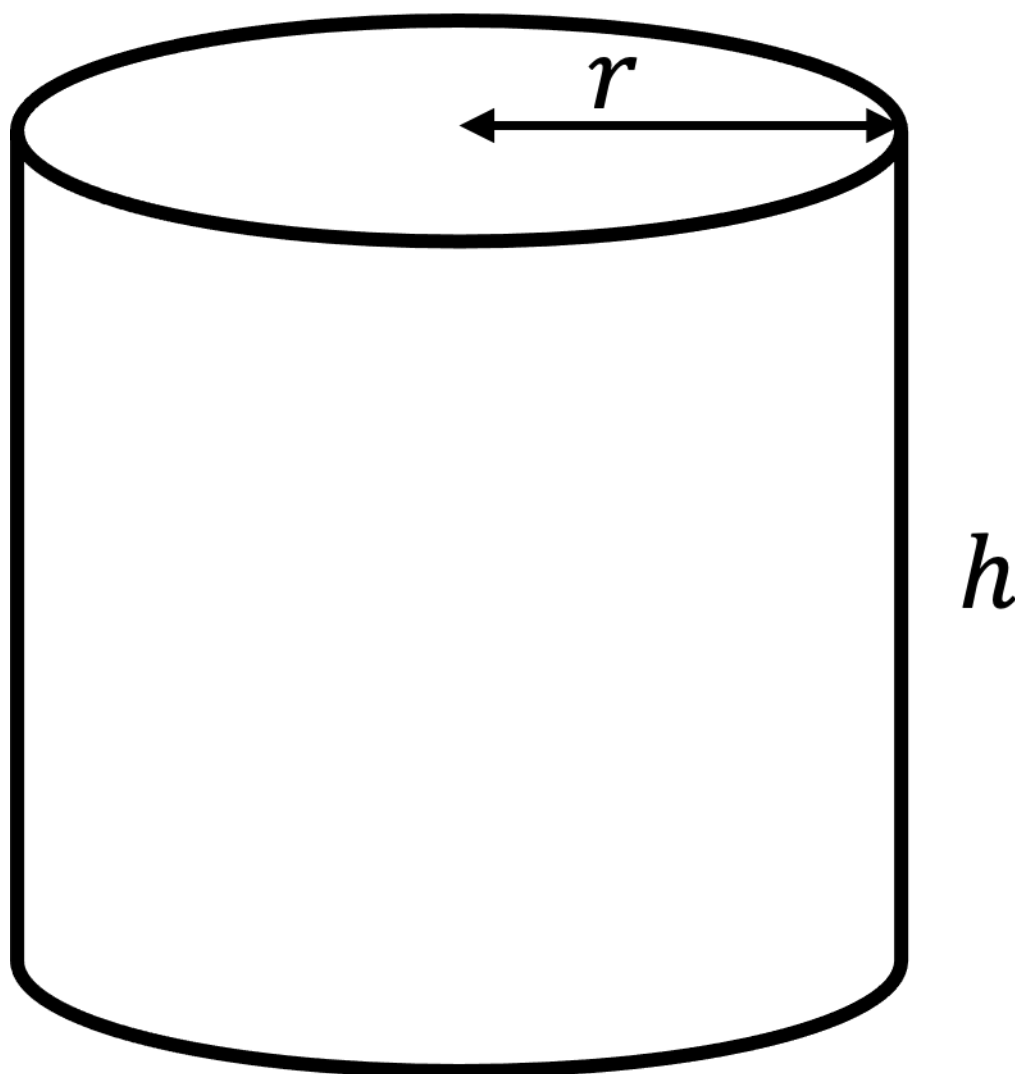


Worked Example

A cylinder is shown.



Your notes



The radius, r , is 8 cm and the height, h , is 20 cm.

Calculate the volume of the cylinder, giving your answer correct to 3 significant figures.

A cylinder is similar to a prism but with a circular base

The volume of any prism, V , is its base area $\times$ height, h , where the base area here is for a circle, πr^2

$$V = \pi r^2 h$$

Substitute $r = 8$ and $h = 20$ into the formula

$$V = \pi \times 8^2 \times 20$$

Work out this value on a calculator

4021.238...

Round the answer to 3 significant figures

4020 cm³



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Your notes

Problem Solving with Volumes

Problem Solving with Volumes

What is problem solving?

- **Problem solving**, usually has two key features:
 - A question is given as a **real-life scenario**
 - eg. The volume of water in a swimming pool...
 - There is usually **more than one topic of maths** you will need in order to answer the question
 - eg. Volume and money

What are common problems that involve volume?

- **Volume** is a commonly used topic of '**real-world**' maths
 - For example, a carton of juice in the shape of a cuboid, a cylindrical tin and a triangular prism chocolate box all involve volume
- Typically, the 'real-world' scenarios also have a **cost**
 - A lot of volume problems also involve **calculations with money**

How do I solve problems involving volume?

- Often the 3D object in a question will **not** be a standard cuboid, cone, sphere, etc.
 - It will likely either be:
 - A **prism** (3D shape with the same cross-section running through it)
 - A **portion** or **fraction** of a standard shape (a hemisphere for example)
 - A **compound object** (an object made up of two or more standard 3D objects)
- If the object is a **prism**, recall that the **volume of a prism** is the **cross-sectional area $\times$ its length**
 - The cross-sectional area may be a **compound 2D shape**
 - For example, an L-shape, or a combination of a rectangle and a triangle
- If the object is a **fraction** of a standard shape, consider the "**full**" **version** of the object and find the **appropriate fraction** of it
 - A hemisphere is half a sphere



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- A **frustum** is a **truncated** (chopped-off) **cone** or **pyramid**
 - The volume of a frustum will be the volume of the smaller cone or pyramid subtracted from the volume of the larger cone or pyramid
- If the object is a **compound object**, find the volumes of the **individual** standard 3D objects and **add** them together
- Problem solving questions could appear on either a **non-calculator paper** or a **calculator paper**



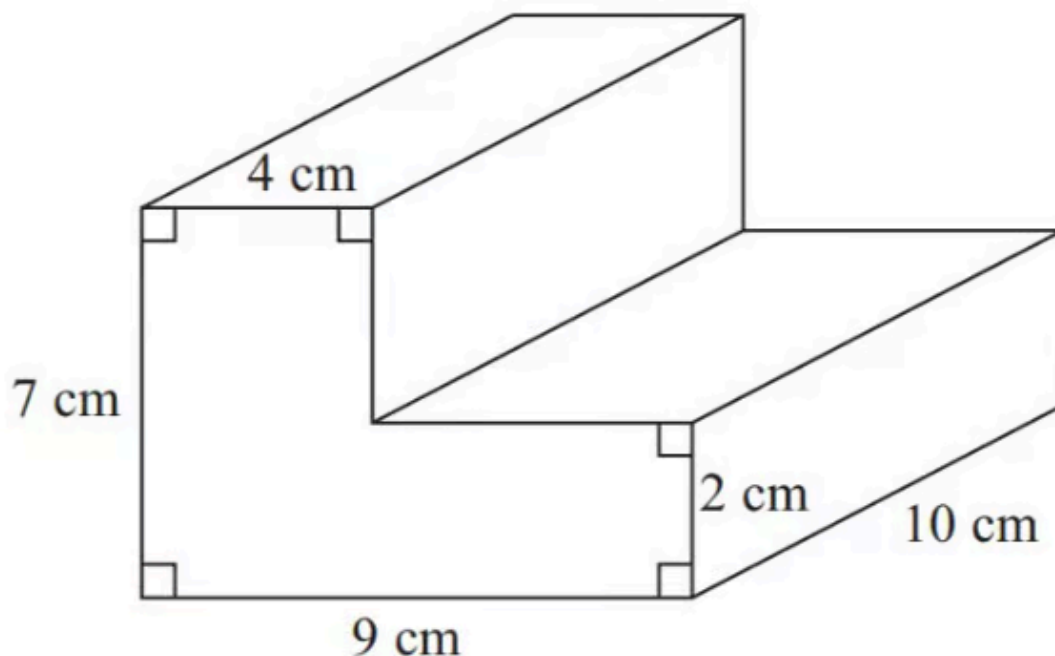
Examiner Tips and Tricks

- Before you start calculating, make a quick note of your plan to tackle the question
 - For example, "Find the area of the triangle and the rectangle, add together, multiply by the length"



Worked Example

The diagram shows a prism.





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Work out the volume of the prism.

The volume is the area of the cross section $\times$ length (10 cm)

Find the area by splitting into a 7×4 and a $(9 - 4) \times 2$ rectangle (or a 9×2 and a $(7 - 2) \times 4$ rectangle)

$$7 \times 4 + (9 - 4) \times 2 = 38 \text{ cm}^2$$

Find the volume (by multiplying 38 by 10)

$$38 \times 10$$

$$380 \text{ cm}^3$$



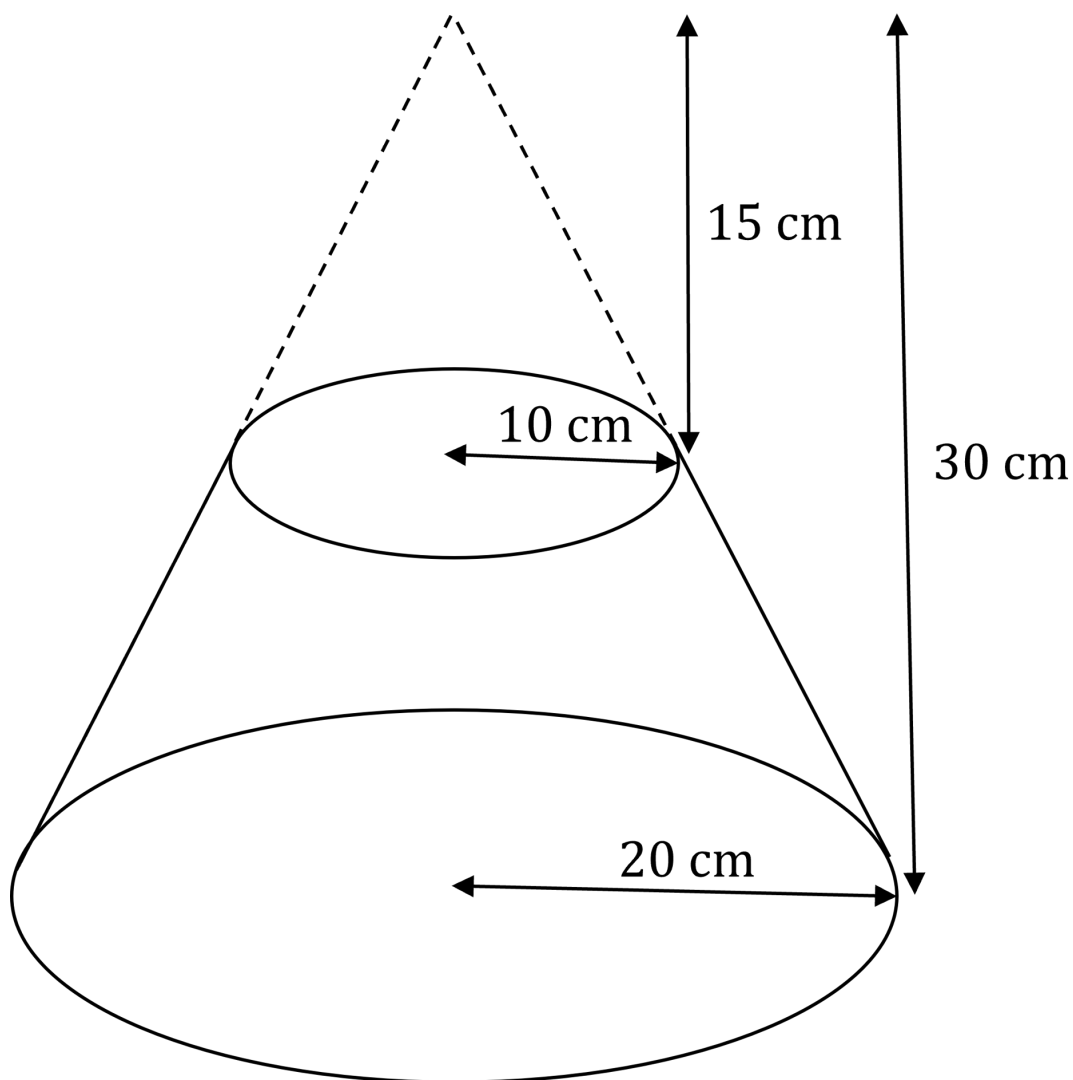
Worked Example

The diagram shows a truncated cone (a frustum).

Using the given dimensions, find the volume of the frustum to 3 significant figures.



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To find the volume of the frustum, find the volume of the larger cone (30 cm tall, with a radius of 20 cm), and subtract the volume of the smaller cone (15 cm tall, with a radius of 10 cm)

Formula for the volume of a cone: $\frac{1}{3} \pi r^2 h$

Calculate the volume of the larger cone

$$V_L = \frac{1}{3} \times \pi \times 20^2 \times 30 = 4000\pi = 12\,566.37061... \text{ cm}^3$$



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Calculate the volume of the smaller cone

$$V_s = \frac{1}{3} \times \pi \times 10^2 \times 15 = 500\pi = 1\,570.796327... \text{ cm}^3$$

Find the difference

$$V_L - V_s = 4000\pi - 500\pi = 3500\pi = 10\,995.57429... \text{ cm}^3$$

Round to 3 significant figures

11 000 cm³



Your notes

Surface Area

Surface Area

What is surface area?

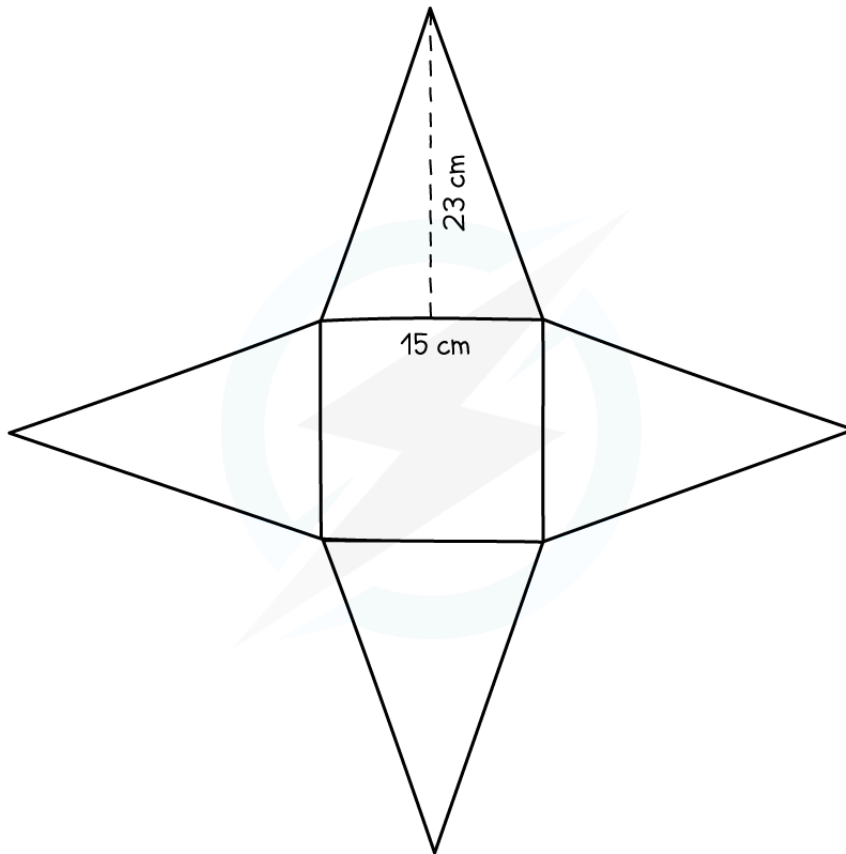
- The **surface area** of a 3D object is the **sum of the areas of all the faces** that make up the shape
 - Area is a 2D idea being applied into a 3D situation
 - A face is one of the flat or curved **surfaces** that make up a 3D object

How do I find the surface area of cubes, cuboids, pyramids, and prisms?

- In cubes, cuboids, polygonal-based pyramids, and polygonal-based prisms (ie. pyramids and prisms whose bases have straight sides), **all the faces are flat**
- The **surface area** is found by
 - calculating the area of each individual flat face
 - adding these areas together
- You should remember the formula for the **area of a rectangle**, but you are given the **area of a triangle** in your exam
- When calculating surface area, it can be helpful to **draw a 2D net for the 3D shape** in question
 - For example, consider a square-based pyramid where the top of the pyramid is directly above the centre of the base
 - Its net will consist of a square base and four identical isosceles triangular faces
 - Calculate the area of a square and the area of each triangle then add them together



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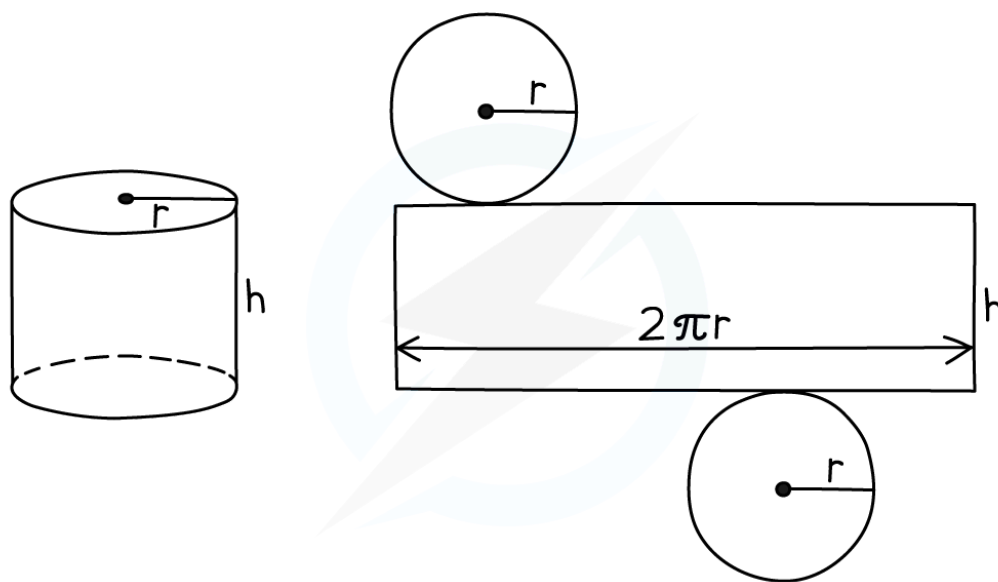


How do I find the surface area of a cylinder?

- A cylinder has **two flat surfaces** (the top and the base) and **one curved surface**
- The **net** of a cylinder consists of **two circles** and a **rectangle**



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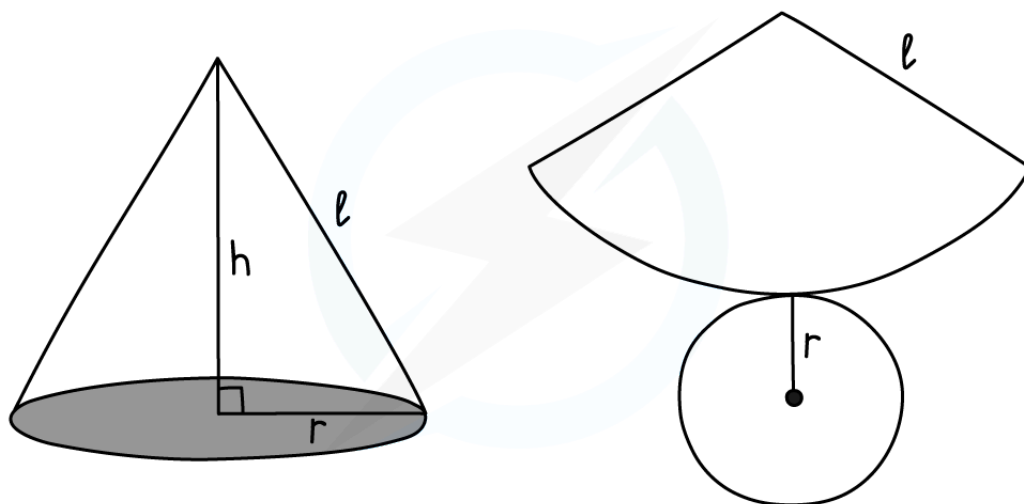
- The **curved surface area** of a cylinder, A , with base radius, r , and height, h , is therefore given by
 - $A = 2\pi r h$
 - This formula is given to you in the exam
- The **total surface area** of a cylinder, A_{Total} , can be found using the formula
 - $A_{Total} = 2\pi r h + 2\pi r^2$
 - This formula is **not** given to you in the exam

How do I find the surface area of a cone?

- A cone has **one flat surface** (the base) and **one curved surface**
- The net of a cone, with radius, r , perpendicular height, h , and sloping edge, (slant height), l , consists of
 - A **circular base**
 - A **sector** with radius, l , and an arc length equal to the circumference of the base



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- The **curved surface area** of a cone, A , with radius, r , perpendicular height, h , and sloping edge, l , can be found using the formula

- $A = \pi r l$

- This formula is given to you in the exam

- The **total surface area** of a cone, A_{Total} , can be found using the formula

- $A_{Total} = \pi r l + \pi r^2$

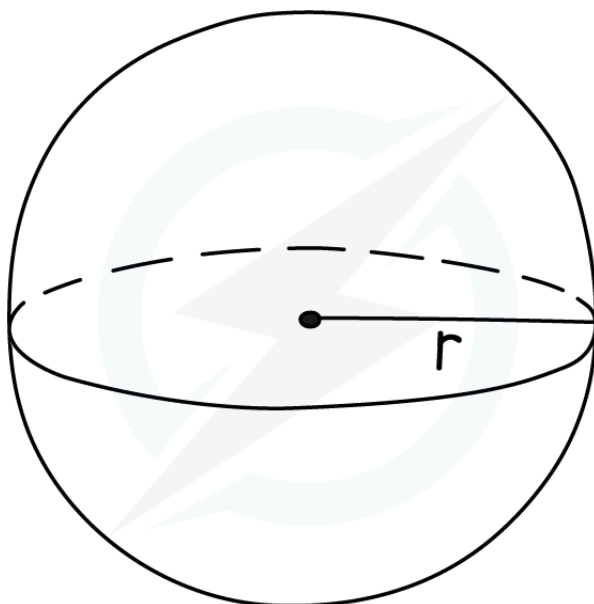
- This formula is **not** given to you in the exam

How do I find the surface area of a sphere?

- A sphere has a **single curved surface**



Your notes



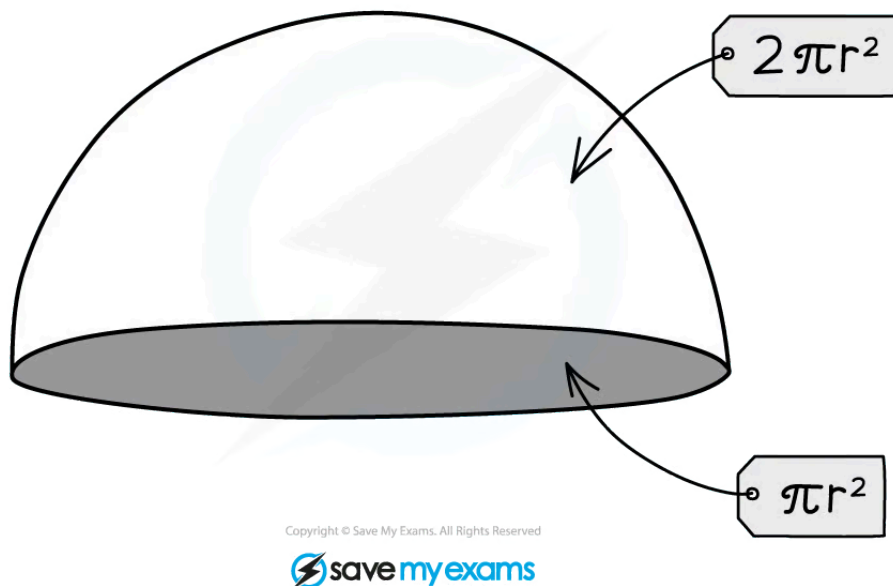
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- The surface area of a sphere, A , with radius, r , can be found using the formula
 - $A = 4\pi r^2$
 - This formula is given to you in the exam
- A **hemisphere** has half the curved surface area of a sphere and the flat circular base



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- The surface area of a hemisphere, A , with radius, r , can be found using the formula

- $A = 2\pi r^2 + \pi r^2$

- This formula is **not** given to you in the exam



Examiner Tips and Tricks

- The **curved surface area** for a cylinder, cone and sphere are given to you in the exam
 - Read the question carefully, you may need to add additional areas, e.g. a base
 - Make you are confident in calculating the areas of rectangles, circles and triangles

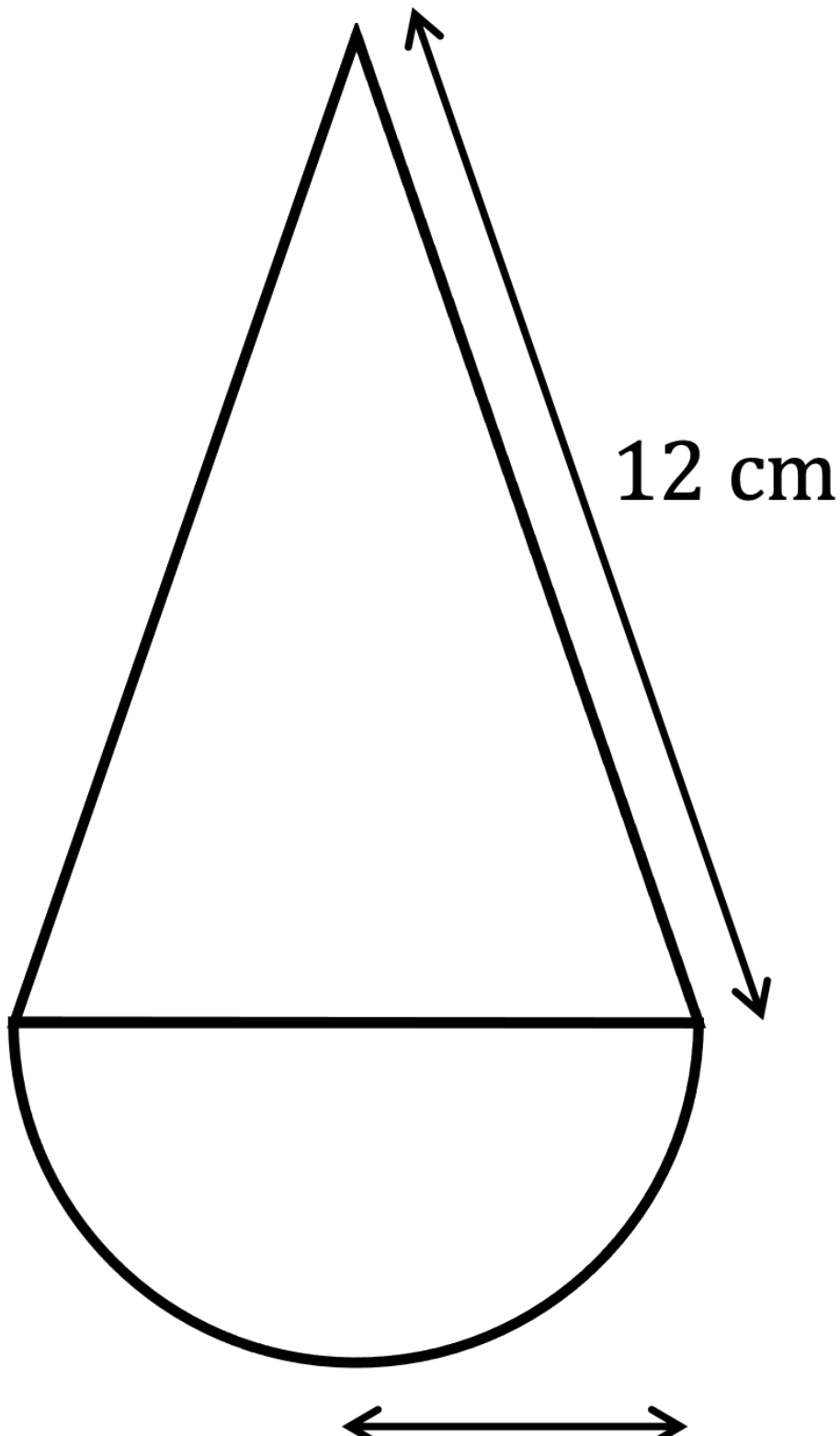


Worked Example

A toy consists of a cone of radius 5 cm and slant height 12 cm placed on top of a hemisphere with the same radius. Find the total surface area of the toy. Give your answer to 3 significant figures.



Your notes





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5 cm

Calculate the curved surface area of the cone using the formula, $A = \pi r l$

$$A_{\text{cone}} = \pi \times 5 \times 12 = 60\pi$$

Calculate the curved surface area of a hemisphere using the formula for the curved surface area of a sphere, $A = 4\pi r^2$ and dividing it by 2

$$A_{\text{hemisphere}} = \frac{4\pi \times 5^2}{2} = 50\pi$$

Add the two areas together to find the total surface area of the toy

$$A_{\text{Total}} = 60\pi + 50\pi = 110\pi$$

Evaluate on your calc and round to 3 significant figures

$$A_{\text{Total}} = 345.57519...$$

346 cm² (3 s.f.)